CS-GY 6063 SOFTWARE ENGINEERING I • GROUP 12-1 • SPRINT 2 DESIGN REVIEW

DegreePath Architecture & System Design

Automated semester planning, prerequisite cycle detection, and minimal conflict explanation

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SECTION 2.1 & 2.2 • SPEAKER 1

Architecture Overview & Component View

Client-Server Monolith

React + TS UI calls six modular Django services over HTTPS JSON.

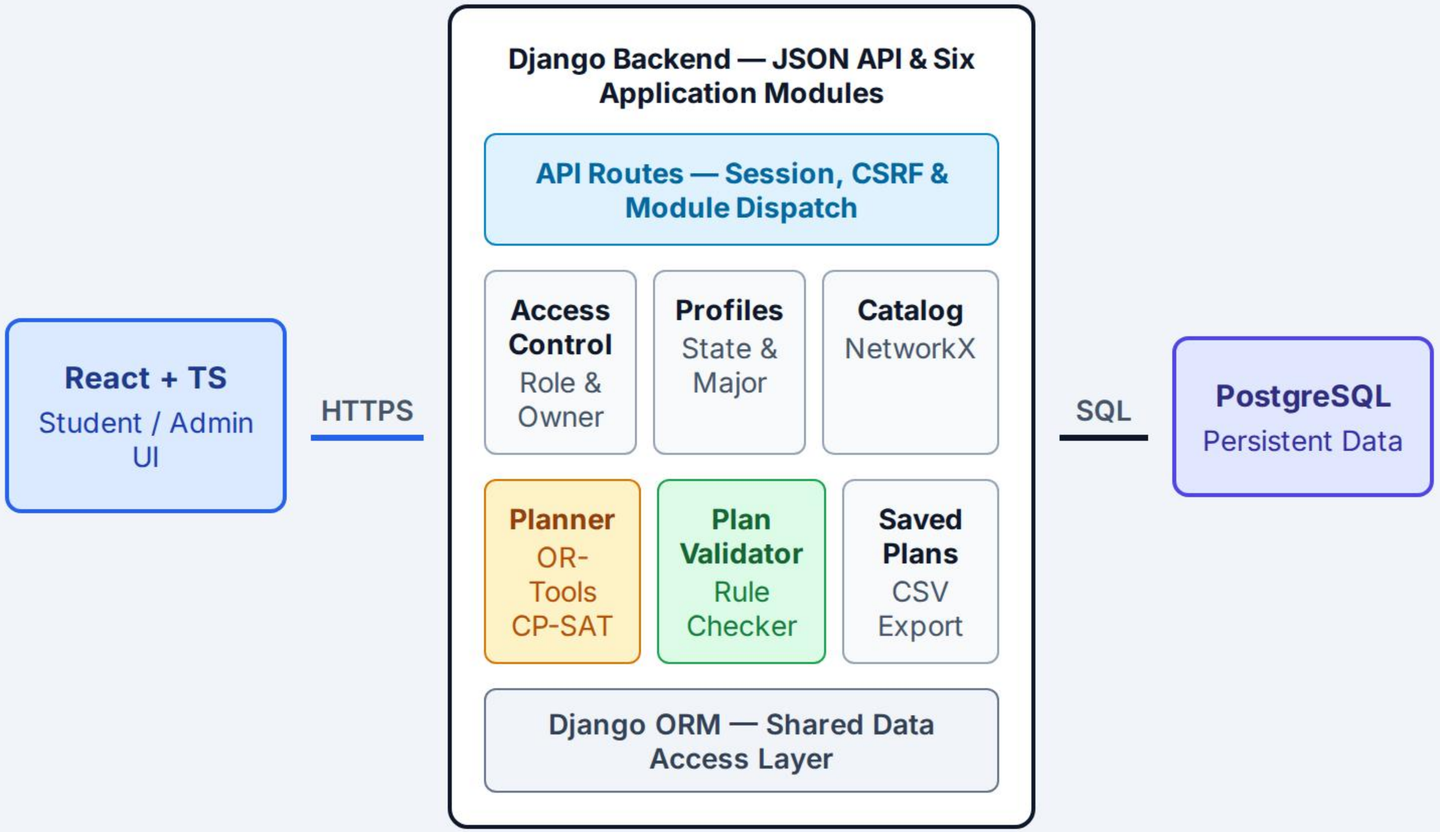
Specialized Core Engines

NetworkX checks import cycles; OR-Tools CP-SAT computes plans.

Independent Validator

Zero solver dependencies; catches modeling bugs before display.

FIGURE 1: DEGREEPATH COMPONENT VIEW



SECTION 2.3 & SECTION 3 • SPEAKER 1

Deployment View & UML Modeling Rationale

FIGURE 2 & D7: SINGLE-ORIGIN PRODUCTION DEPLOYMENT PIPELINE



Design View	Chosen Notation	Engineering Justification (Section 3)
Domain Data Model	Crow's-Foot ERD	Maps directly to PostgreSQL tables with zero inheritance; shows PK, FK, UK clearly.
Generation & Import	UML Sequence	Crosses five modules with alt fragments for auth, transactions, and solver outcomes.
Conflict Explainer	UML Activity	Internal algorithmic loop inside one module; lifelines add zero value.
Saved Plan Lifecycle	UML State Machine	Plan is the only stateful entity transitioning across Unsaved, Current, and Out of Date.

SECTION 4.1 – 4.5 • SPEAKER 2

Data Design & Crow's-Foot Entity-Relationship Model

Figure 3: Persistent Entities Grouped by Domain (13 Tables)

ACCOUNTS (1:1 / M:N)

USER (id PK, email UK, role)
STUDENT_PROFILE (UUID PK, credit_limit)
COMPLETED_COURSE (profile_id, course_id)

DEGREE RULES (1:N / M:N)

DEGREE_PROGRAM (id PK, name UK, kind)
REQUIREMENT (type, choose_n, min_credits)
REQUIREMENT_COURSE (req_id, course_id)

CATALOG & CNF PREREQS

COURSE (code UK, credits, manual_review)
COURSE_OFFERING (season ∈ Fall, Spring)
PREREQ_CLAUSE + **COREQUISITE**

PLANS & BOOKKEEPING

PLAN (UUID PK, profile_ver, catalog_ver)
PLAN_ENTRY (term_index, counted_for FK)
CATALOG_IMPORT & **CATALOG_STATE**

Normalized CNF Prereqs

ANDed clauses of ORed courses with strict foreign-key integrity.

Canonical Corequisites

CHECK course_id < coreq_course_id prevents duplicate pairs.

Transient Conflict Set

Recomputed on demand across 9 ConstraintRef types; never stored.

SECTION 5.1 – 5.5 • SPEAKER 2

REST API Interface & 4-Gate Access Pipeline

GATE 1: SESSION
Cookie < 24h old else **401**

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GATE 2: ROLE
Student / Admin else **403**

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GATE 3: EXISTENCE
UUID lookup else **404**

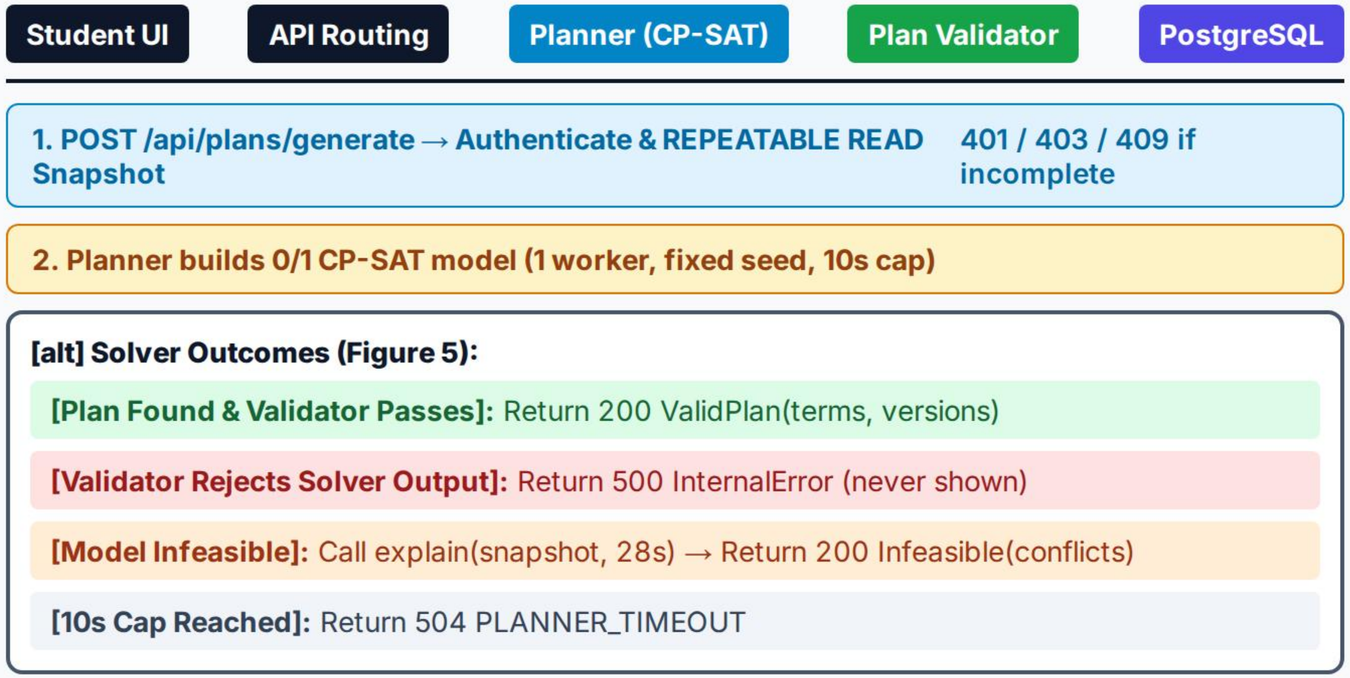
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GATE 4: OWNERSHIP
Caller owns UUID else **403**

Endpoint	Caller	Success / Validation Behavior	Error Statuses	Stories
`PUT /api/profiles/{id}`	Owner	Atomic update; increments profile_version	**400, 401, 403, 404**	US-02, US-03
`POST /api/admin/catalog-imports`	Admin	201 atomic upsert; increments catalog_version	**400 Cycle/Row, 403**	US-05–US-07
`POST /api/plans/generate`	Student	200 valid terms OR 200 infeasible + conflict set	**409 Incomplete, 504**	US-09, US-12
`POST /api/plans`	Student	Reloads snapshot & re-runs validator before save	**400 Invalid, 409 Stale**	US-10, NFR-4
`GET /api/plans/{id}/export.csv`	Owner	200 text/csv ordered by term_index	**409 Out of Date, 403**	US-13, NFR-1

SECTION 6.1 • FIGURES 4 & 5 • SPEAKER 3

Plan Generation & Validation Sequence



Snapshot Isolation

Single REPEATABLE READ transaction prevents mid-solve catalog drift.

Deterministic Solve

Single worker and fixed seed guarantee identical plans (US-09 AC4).

Safety Net Validator

Verifies prerequisites, coreqs, caps, and pool allocations independently.

SECTION 6.2 & 6.3 • FIGURES 6 & 7 • SPEAKER 3

Catalog Import & Minimal Conflict Explanation

FIGURE 6: ALL-OR-NOTHING CATALOG IMPORT

1. Lock CatalogState FOR UPDATE
Serializes concurrent admin uploads cleanly.

2. Parse CNF & Merge Corequisite Nodes
NetworkX detects A → B → C → A and coreq loops.

3. Savepoint Upsert or Rollback
Any error rolls back writes and logs rejected report.

FIGURE 7: CONFLICT EXPLANATION ACTIVITY FLOW

1. Guard 9 Explainable Constraint Types
Attach assumption literals; extract sufficient core.

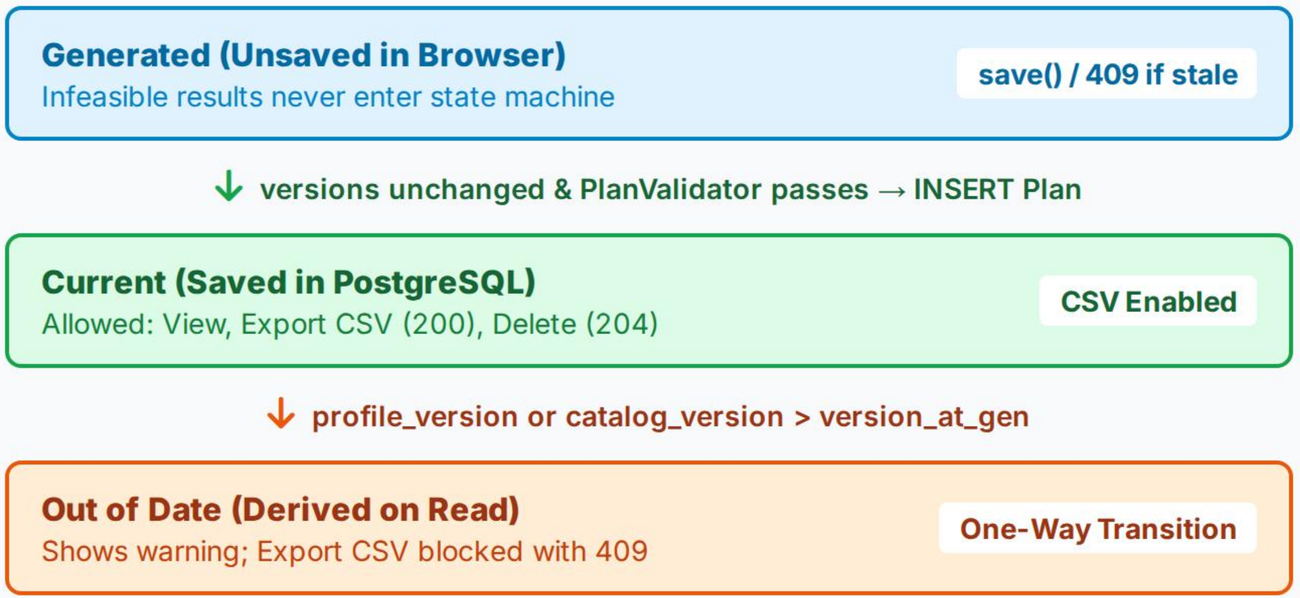
2. Deletion-Based Minimization Loop
Test core minus c: if still infeasible, remove c permanently.

3. Enforce 28s Request Deadline
Sets possibly_non_minimal = true if budget expires.

SECTION 6.4 • FIGURE 8 • SPEAKER 4

Plan Lifecycle State Machine & Staleness Rules

FIGURE 8: UML STATE MACHINE FOR PLAN ENTITY



Zero Browser Trust

Save reloads the snapshot and re-validates every course entry.

Monotonic Versions

Versions only increase; stale plans never revert to Current.

Cascading Delete

Confirmed delete removes Plan and PlanEntry rows; URL returns 404.

SECTION 7 • ARCHITECTURE DECISION RECORDS • SPEAKER 4

Four Key Architectural Decision Records (ADRs)

ADR-01 • HAN YANG

Django Sessions Over FastAPI + JWT

Eliminates custom token refresh logic via 24h HttpOnly cookies and CSRF protection.

ADR-02 • HANYUE LIAO

Transactional All-or-Nothing Imports

Wraps catalog and requirement uploads in atomic transactions without staging-table overhead.

ADR-03 • ACHYUTHAN SIVASANKAR

Google OR-Tools CP-SAT Solver

Models 0/1 course-term variables directly and extracts minimal conflict cores under 30s.

ADR-04 • OMKAR GADUTE

Normalized CNF Prerequisite Tables

Enforces foreign keys at import time and maps 1-to-1 to CP-SAT clauses and NetworkX edges.

SECTIONS 8, 9 & 10 • SPEAKER 5

NFR Verification Matrix & Design Discussion Log

NFR	Design Mechanism	Automated Verification Test
NFR-1 Permissions	require_role, require_owner, random UUIDs (DR-4)	Parametrized API tests asserting 401/403 + CI route-table test
NFR-2 Security	Argon2id hashes, 24h DB sessions, HttpOnly/CSRF	DB hash inspection (argon2$), injected clock expiry test
NFR-3 Performance	1-worker CP-SAT (10s cap) + 28s shared explain budget (DR-8)	20 timed runs on pilot major asserting p95 ≤ 10s in CI
NFR-4 Integrity	transaction.atomic imports + independent PlanValidator (DR-3)	Row-count rollback test + 50-scenario validator suite (100% pass)
NFR-5 & NFR-6	Allow-list LoggingFilter + Neon PostgreSQL & pg_dump	Canary password/course log-scan + container restart diff script

DR-7: Pool Allocation Verification
Client sends counted_for on save; validator verifies exact pool match.

DR-9: MANUAL_REVIEW Constraint
9th explainable ConstraintRef type alerts students to seek permission.

Sprint 1 Revisions (Section 9)
All 14 MVP stories accepted unchanged; minor Sprint 1 typos fixed.

SPRINT 1 REQUIREMENTS BACKLOG • WORK ESTIMATES • SPEAKER 5

User Story T-Shirt Size Estimates (14 MVP + 4 Stretch)

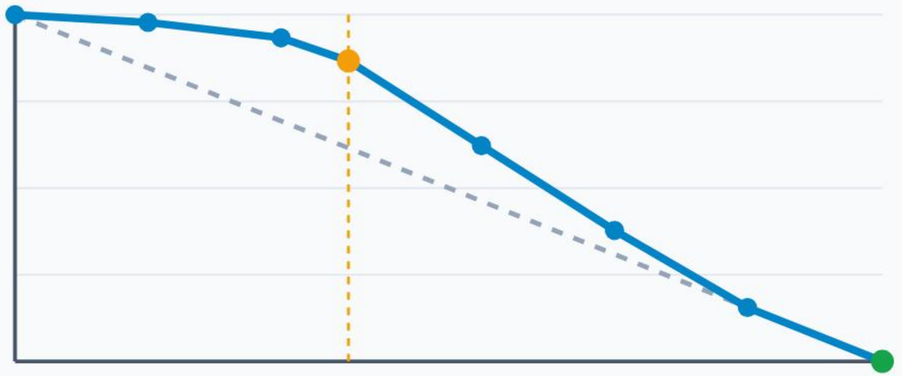
Story	MVP / Stretch Requirement	Owner	Size
US-01	Register & sign in (Argon2)	Maitri	**M**
US-02	Enter profile, major, credit cap	Maitri	**M**
US-03	Update profile & flag stale plans	Han	**M**
US-04	Catalog search & rule display	Hanyue	**S**
US-05	Admin CSV/JSON course import	Han	**M**
US-06	AND/OR CNF prereqs & coreqs	Omkar	**L**
US-07	Prereq/coreq cycle detection	Achyuthan	**L**
US-08	Import degree pools/thresholds	Hanyue	**M**
US-14	Stretch #1: What-if comparison	Omkar	**L**

Story	MVP / Stretch Requirement	Owner	Size
US-09	Generate semester plan (CP-SAT)	Achyuthan	**XL**
US-10	Save & reopen validated plans	Omkar	**M**
US-11	Delete saved plan permanently	Hanyue	**S**
US-12	Minimal conflict set explanation	Achyuthan	**XL**
US-13	Export current valid plan as CSV	Maitri	**S**
US-18	Satisfy major pools & thresholds	Achyuthan	**L**
US-15	Stretch #2: Single-change fixes	Han	**L**
US-16	Stretch #3: Calendar .ics export	Hanyue	**S**
US-17	Stretch #4: Declare minor rules	Maitri	**M**

SPRINT 2 EXECUTION METRICS • 7 OCT – 20 OCT 2026 (2-WEEK SPRINT)

Sprint 2 Design Burndown Chart & Milestones

Remaining Design Effort (45 Story Points Total) -- Ideal Burndown — Actual Burndown



7 Oct (Start) 10 Oct (Spikes) 12 Oct (Merge) 15 Oct (ADRs) 18 Oct (UMLs) 20 Oct (Done)

Oct 7–11: Individual Research & Spikes

Split tasks across 5 members; prototyped CP-SAT, ERD tables, and API routes independently.

Oct 12–16: Consolidation & Core ADRs

Started merging sections into shared doc; locked in DR-1 to DR-6 and unified module names.

Oct 17–20: Peer Review & Final Polish

Resolved review comments (DR-7 to DR-15), verified traceability tables, and completed all 8 diagrams.